

[ELECTROIC ANALYSIS]

[DAPP Program]

Abstract

[final project of electronics Analysis]

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About Project :

- 1- I analyzed two files, a text file for reviews and a csv file for daily sales using the following tools: Python, Excel, Power BI
- 2- Create a database using SQL using Xampp
- 3- Link these different sources to Power BI
- 4- Represent the data in Power BI

1- Cleaning the data:

Clean Daily sales file:

Python:

Firstly I used Python to handle missing values , changed the data type to the appropriate data type, and then exported the file to Power bi to continue the cleaning process

```
import pandas as pd

# Load the data
df = pd.read_csv('/content/Updated_sales.csv')

# Display the first few rows of the dataframe
print("Original Data:")
print(df.head())

# 1. Check for missing values
print("\nMissing Values:")
print(df.isnull().sum())

# 2. Remove rows with missing values (if any)
df.dropna(inplace=True)

# 3. Remove duplicates
df.drop_duplicates(inplace=True)

# 4. Standardize the 'Order Date' format
df['Order Date'] = pd.to_datetime(df['Order Date'], errors='coerce')

# 5. Convert 'Quantity Ordered' and 'Price Each' to numeric (if not already)
df['Quantity Ordered'] = pd.to_numeric(df['Quantity Ordered'], errors='coerce')
df['Price Each'] = pd.to_numeric(df['Price Each'], errors='coerce')

# 6. Remove any rows that became NaT/NaN after conversions
df.dropna(inplace=True)

# 7. Check the data types
print("\nData Types After Cleaning")
```

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```
# 7. Check the data types
print("\nData Types After Cleaning:")
print(df.dtypes)

# 8. Reset index after cleaning
df.reset_index(drop=True, inplace=True)

# 9. Save cleaned data to a new CSV file
df.to_csv('cleaned_data.csv', index=False)

# Display the cleaned data
print("\nCleaned Data:")
print(df.head())
```

Original Data:

	Order ID	Product	Quantity Ordered	Price Each	\
0	176558	USB-C Charging Cable	2	11.95	
1	NaN	NaN	NaN	NaN	
2	176559	Bose SoundSport Headphones	1	99.99	
3	176560	Google Phone	1	600	
4	176560	Wired Headphones	1	11.99	

	Order Date	Purchase Address
0	04/19/19 08:46	917 1st St, Dallas, TX 75001
1	NaN	NaN
2	04/07/19 22:30	682 Chestnut St, Boston, MA 02215
3	04/12/19 14:38	669 Spruce St, Los Angeles, CA 90001
4	04/12/19 14:38	669 Spruce St, Los Angeles, CA 90001

Missing Values:

Order ID	87
Product	87
Quantity Ordered	87

2-Powe bi :

Separated the Address and data filed into 4 columns to help in the analysis

And then delete duplicate columns after these processes like Order data columns

Change order_ID into string data type

	1 st Street	2 nd City	3 rd State	4 th Zip code	5 th Month Name	6 th Year	7 th Week of Year
1	917 1st St	Dallas	TX	75001	Apr	2019	
2	682 Chestnut St	Boston	MA	2215	Apr	2019	
3	669 Spruce St	Los Angeles	CA	90001	Apr	2019	
4	333 8th St	Los Angeles	CA	90001	Apr	2019	
5	381 Wilson St	San Francisco	CA	94016	Apr	2019	
6	668 Center St	Seattle	WA	98101	Apr	2019	
7	790 Ridge St	Atlanta	GA	30301	Apr	2019	
8	915 Willow St	San Francisco	CA	94016	Apr	2019	

The file is ready to analysis

2- cleaning txt file:

I used Python to analyze the file based on Positive Keywords:

```
positive_words = [  
    "good", "appreciate", "thrilled", "thank", "easier", "easy", "great",  
    "love", "long", "better", "quick", "quickly", "best", "enjoy", "enjoyed", "nice"  
]
```

```
import pandas as pd  
  
# قائمة الكلمات الإيجابية  
positive_words = [  
    "good", "appreciate", "thrilled", "thank", "easier", "easy", "great",  
    "love", "long", "better", "quick", "quickly", "best", "enjoy", "enjoyed", "nice"  
]  
  
# قراءة ملف آراء العملاء (يجب تعديل المسار ليتناسب مع ملفك)  
# حيث كل سطر يحتوي على نص الرأي متبوعاً بالتاب ثم رمز العميل  
with open('/content/e-revuiw.txt', 'r') as file:  
    lines = file.readlines()  
  
# إنشاء قائمة لتخزين البيانات المنظمة  
data = []  
  
# معالجة كل سطر من الملف  
for line in lines:  
    # تقسيم السطر إلى نص ورمز العميل  
    review_text, _ = line.split('\t')  
  
    # تحليل النص لتحديد ما إذا كان إيجابياً أو سلبياً  
    is_positive = 0 # افتراض أن الرأي سلبي  
    for word in positive_words:  
        if word in review_text and f"not {word}" not in review_text:  
            is_positive = 1  
            break
```

```
        break  
  
# في النص 'Product: ' افتراض أن اسم المنتج يأتي بعد كلمة  
# وإذا لم يكن موجوداً، يتم تركه فارغاً  
product_name = ""  
if "Product: " in review_text:  
    product_name = review_text.split("Product: ")[1].split()[0]  
  
# إضافة السطر المنظم إلى القائمة  
data.append([product_name, is_positive])  
  
# من البيانات المنظمة إنشاء DataFrame  
df = pd.DataFrame(data, columns=["ProductName", "Sentiment"])  
  
# عرض الجدول  
print(df)  
  
# حفظ الجدول في ملف CSV  
df.to_csv('customer_reviews_analysis.csv', index=False)
```

Sample:

	A	B	C
1	ProductName	Sentiment	
2	USB-C Charging C	1	
3	Bose SoundSport I	1	
4	Google Phone	1	
5	Wired Headphone	1	
6	Wired Headphone	1	
7	USB-C Charging C	1	
8	Bose SoundSport I	0	
9	USB-C Charging C	1	

2- Create database:

Using Xampp I create database product

The columns:

Payment_method: credit card ,cash ,e-wallet

Returned : Yes,No

Customer_Rating: in range 1 to 10

create table product (

Payment_method varchar(23),

Returned char(3),

Customer_Rating int

)

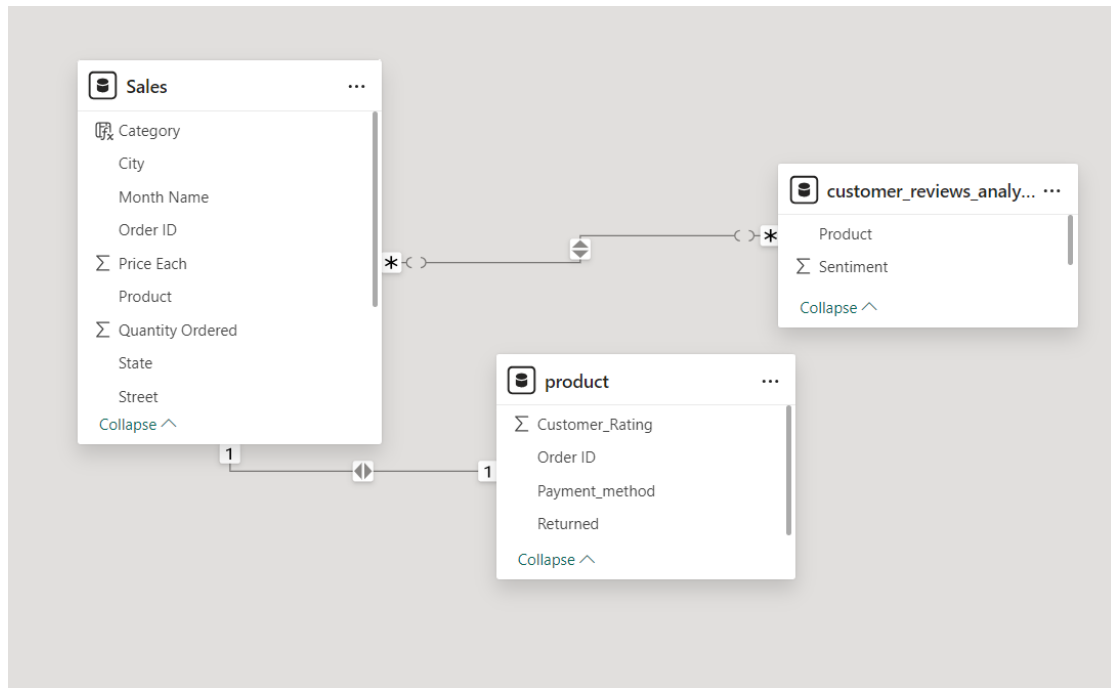
Fill it randomly with 29018 value



3- Connect SQL database and files in Power BI:

Sales and Product : where Order_ID is the primary key

Sales and customer_reviews-analysis : Product is the primary key



Creating Dashboard:



Overall Performance

The dashboards present a snapshot of an electronics business's performance across various dimensions. Key indicators suggest

Strong Sales: Total sales are substantial, with a significant contribution from smartphones and laptops

Geographical Variation: Sales vary across cities, San Francisco being top performers

Seasonal Trends: Sales fluctuate throughout the year, with potential peaks during specific months or periods

Product Popularity: iPhones and MacBooks are top-selling products, followed by other laptops and smartphones

Customer Sentiment: While there's a relatively low negative sentiment count, it's an area to monitor for potential issues

Return Rate: The return rate for specific product subcategories (e.g., batteries) seems relatively high, indicating potential quality or customer satisfaction issues

key Areas for Focus:

Based on the available data, the following areas deserve closer attention

□ Product Performance:

Analyze sales performance by product category and subcategory to identify best-sellers and underperformers. •

Investigate the reasons for high return rates in certain product categories (e.g., batteries). •

Consider product lifecycle management and potential product line extensions. •

□ Customer Analysis:

Deepen the understanding of customer sentiment by analyzing textual data and identifying common complaints or praises. •

Explore customer demographics and preferences to tailor marketing and product offerings. •

Implement customer loyalty programs to encourage repeat business. •

•

□ Geographical Analysis:

Identify factors contributing to sales variations across cities (e.g., competition, market size, economic conditions). •

Explore opportunities for expansion or increased market penetration in underperforming regions. •

□ Sales and Marketing Optimization:

Analyze sales trends by month and week to identify seasonal patterns and optimize marketing campaigns accordingly. •

Evaluate the effectiveness of different marketing channels and allocate resources accordingly. •

Explore opportunities for cross-selling and upselling based on customer purchase behavior. •

Using DAX:

to categorize the product column into (Cables, Headphone, Televisions, Smatphone, laptops, Batteries", Home Appliances, Miscellaneous) to make more informative dashboards

1

Category =

2

SWITCH(

3

TRUE(),

4

Sales[Product] IN {"USB-C Charging Cable", "Lightning Charging Cable"}, "Cables ",

5

Sales[Product] IN {"Wired Headphones", "Bose SoundSport Headphones", "Apple AirPods Headphones"}, "Headphones",

6

Sales[Product] IN {"Vareebadd Phone", "Google Phone", "iPhone"}, "Smartphones",

7

Sales[Product] IN {"Flatscreen TV", "20in Monitor", "27in 4K Gaming Monitor", "27in FHD Monitor", "34in Ultrawide Monitor"}, "Televisions ",

8

Sales[Product] IN {"MacBook Pro Laptop", "ThinkPad Laptop"}, "Laptops",

9

Sales[Product] IN {"LG Dryer", "LG Washing Machine"}, "Home Appliances",

10

Sales[Product] IN {"AA Batteries (4-pack)", "AAA Batteries (4-pack)"}, "Batteries",

11

"Miscellaneous" // Default category if no match is found

12

)

Product	Quantity Ordered	Price Each	Total	Street	City	State	Zip code	Category	Month Name	Year	Week of Year
Google Phone	1	600	600	770 Maple St	San Francisco	CA	94016	Smartphones	Apr	2019	15
Lightning Charging Cable	1	14.95	14.95	289 4th St	San Francisco	CA	94016	Cables	Apr	2019	15
Apple AirPods Headphones	1	150	150	872 North St	San Francisco	CA	94016	Headphones	Apr	2019	15
Bose SoundSport Headphones	1	99.99	99.99	60 Maple St	San Francisco	CA	94016	Headphones	Apr	2019	15
34in Ultrawide Monitor	1	379.99	379.99	13 Cherry St	San Francisco	CA	94016	Televisions	Apr	2019	15
Bose SoundSport Headphones	1	99.99	99.99	364 Ridge St	San Francisco	CA	94016	Headphones	Apr	2019	15
ThinkPad Laptop	1	999.99	999.99	990 9th St	San Francisco	CA	94016	Laptops	Apr	2019	15
Flatscreen TV	1	300	300	26 Hickory St	San Francisco	CA	94016	Televisions	Apr	2019	15
USB-C Charging Cable	1	11.95	11.95	839 Elm St	San Francisco	CA	94016	Cables	Apr	2019	15
Flatscreen TV	1	300	300	353 Dogwood St	San Francisco	CA	94016	Televisions	Apr	2019	15
34in Ultrawide Monitor	1	379.99	379.99	243 Hill St	San Francisco	CA	94016	Televisions	Apr	2019	15
Flatscreen TV	1	300	300	799 Sunset St	San Francisco	CA	94016	Televisions	Apr	2019	15
Apple AirPods Headphones	1	150	150	664 Church St	San Francisco	CA	94016	Headphones	Apr	2019	15
Apple AirPods Headphones	1	150	150	914 West St	San Francisco	CA	94016	Headphones	Apr	2019	15
Lightning Charging Cable	1	14.95	14.95	793 6th St	San Francisco	CA	94016	Cables	Apr	2019	15
Vareebadd Phone	1	400	400	891 Jefferson St	San Francisco	CA	94016	Smartphones	Apr	2019	15
Google Phone	1	600	600	417 13th St	San Francisco	CA	94016	Smartphones	Apr	2019	15

Make sentiment text to analysis in pie chart make it more clear:

<pre> 1 SentimentText = IF([Sentiment] = 1, "Positive", "Negative") </pre>		
Sentiment		SentimentText

Exextart first 3 characters from Month name:

	Month Name	Year
nes	Apr	20
	Apr	20
nes	Apr	20
nes	Apr	20
is	Apr	20
nes	Apr	20